# WAIPS at the NTCIR-19 FEHU: Label-Aware Prompt Ensembling and Frequency-Gated Heterogeneous Fusion for Human-Value Recognition

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## Abstract

We present the three official runs submitted to NTCIR-19 FEHU Task 1, which asks systems to recognize actor-conditioned human values in news and to predict direction for fine-grained values. Development experiments showed a recurring trade-off: conservative prompt ensembles preserve direction but miss infrequent values, whereas recall-oriented methods recover more labels but add false positives. WAIPS-01 uses label-wise development analysis to determine which labels a targeted prompt may add to a conservative majority. For WAIPS-03, we augment difficult directed labels and tune DeBERTa-v3-base with LoRA and masked partial-label supervision. WAIPS-03 implements our Frequency-Gated Heterogeneous Fusion (FGHF), which combines the resulting encoder probabilities with heterogeneous prompt jurors through calibrated veto and addition rules, followed by focused direction verification. WAIPS-03 scores 0.5808 Micro-F1 and 0.3952 Macro-F1 on official Task 1b, the highest Macro-F1 among participant runs. WAIPS-01 has the lowest participant Direction Reverse Rate (DRR), 0.0345. Targeted augmentation improves score separation for 18 low-frequency labels and supplies auxiliary encoder evidence, while E6b provides direct tail recall. These components jointly contribute to the macro-oriented behavior of the final system.

## Keywords

human values, multi-label classification, long-tail learning, data augmentation, prompt ensemble, calibrated fusion

## Team Name

WAIPS

## Subtasks

Task 1a and Task 1b

## 1 Introduction

Human values are often expressed indirectly in news [3]. A report may describe the same decision as protecting public safety for one actor and restricting freedom for another. Recognizing a value therefore requires more than locating a topical phrase: the system must identify whose action is being judged, which value it concerns, and whether the action aligns with or contradicts that value. FEHU Task 1 defines these decisions explicitly [1]. Task 1a asks for directionless Level-2 values for each article–actor pair, while Task 1b evaluates directed Level-1 values. We refer readers to the Task Overview by Wang et al. for the collection construction, taxonomy, and official evaluation protocol [1, 2].

Three properties of the task shaped our design. The output space is strongly imbalanced. A small group of values appears throughout the training set, while many directed Level-1 classes have only a few examples. This resembles the long-tail behavior studied in imbalanced classification [9, 10]. A system can therefore obtain reasonable Micro-F1 while never predicting several valid labels. Increasing recall is also unusually expensive: every extra value creates another opportunity for both a false positive and a direction reversal. The available models also make different errors. Prompt variants make complementary errors, and the encoder can expose weak tail-label evidence that is not reliable enough to use as a final decision by itself. Our use of agreement as evidence is related to consistency-based prompting [8], although our rule is an exact-tuple, label-aware ensemble rather than chain-of-thought self-consistency.

The components are not combined in a single voting rule; each has a distinct role. WAIPS-01, developed by Liu, starts with a conservative prompt majority and selectively restores predictions for labels on which a targeted prompt performed better on development data. A separate augmentation track diagnoses difficult directed labels, generates focused training cases, and trains a parameter-efficient DeBERTa classifier to supply continuous evidence. WAIPS-03 uses our Frequency-Gated Heterogeneous Fusion (FGHF; internal E7-G4 configuration), with E6b as a recall-oriented primary generator and other prompt runs plus the augmented encoder acting as jurors. A calibrated gate removes weak frequent-class predictions and admits a small number of supported additions. Direction is then checked only when jurors explicitly disagree.

We interpret the results according to these roles. E6b supplies much of the direct tail recall, the DeBERTa track improves representation of selected low-frequency labels and contributes an auxiliary probability signal, and the fusion layer controls the precision cost of combining them. We report both F1 and DRR throughout because the official runs occupy different points on that trade-off.

## 2 Task Setting and Submitted Runs

For each actor listed in an article, Task 1b requires a set of (Level-1 value, direction) pairs. We treat the two directions of each of 54 Level-1 values as 108 atomic classes. Task 1a separately evaluates directionless Level-2 values; the official task definition does not require a system to derive one subtask's output from

the other [1]. Each submitted run therefore uses the run-specific Task-1a construction described below. Development experiments use the official 177-article development set. The frozen inference pipelines were then applied to 175 test articles containing 479 released article–actor pairs; no test labels or Task-2 annotations were used.

We follow the official evaluator and the Task Overview [1] in treating Micro-F1 as the primary ranking measure. Macro-F1 is averaged over labels that occur at least once in the test gold data, making it a useful but support-dependent view of tail behavior. DRR counts opposite-direction predictions for gold actor–value pairs with a unique direction. A missed value does not count as a reversal, so a low DRR must be read together with F1 rather than as a standalone accuracy measure.

The three submissions use different designs. WAIPS-01 is Liu's label-aware prompt ensemble, described in Section 3. WAIPS-02 uses a three-way majority vote over DeepSeek Pro prompt variants and uses a Flash prediction, when available, to arbitrate direction. WAIPS-03 is the submitted FGHF run described in Sections 5 and 6. This paper focuses on WAIPS-01 and WAIPS-03 because they represent the two main methodological lines; WAIPS-02 is retained as a prompt-only reference point.

**Table 1: The three submitted WAIPS runs.**

| Run | Short description |
|---|---|
| WAIPS-01 | Label-aware prompt ensemble; direction-stable selective recovery. |
| WAIPS-02 | Three-way prompt majority with direction arbitration. |
| WAIPS-03 | Frequency-gated heterogeneous fusion. |

WAIPS-02 is an intermediate prompt-only system. One DeepSeek Pro prompt is the baseline, one is compact and strict, and one uses targeted balanced few-shot examples; all run at temperature zero. A label is retained when at least two runs emit the same actor, value, and direction. The Flash run is then consulted only for direction: when it predicts the same actor and value with the opposite direction, the majority direction is replaced. Task 1a is deterministically projected from the resulting Task-1b set. Unlike FGHF, this run has no learned calibrator, encoder features, frequency gate, or per-cell QA call.

## 3 Label-Aware Prompt Ensembling (WAIPS-01)

### 3.1 Method motivation and prompt-only design

WAIPS-01 is Liu's standalone prompt-only submission and was developed separately from the later FGHF system. Development experiments showed that prompt variants can have complementary error profiles. A strict evidence requirement makes the system conservative and helps avoid unsupported values and direction reversals, but it also suppresses true labels. A targeted prompt recovers difficult values but adds false positives. We formulate ensembling as a label-specific selection problem, rather than treating every prompt prediction equally. This development-set analysis is related to label-wise long-tail diagnosis [9, 10], but the fixed directed-label set and exact-tuple rule were determined by our own FEHU development experiments. Transformer ensembles were also effective in the earlier ValueEval shared task [6]; our ensemble instead operates on structured prompt outputs and uses label-specific recovery.

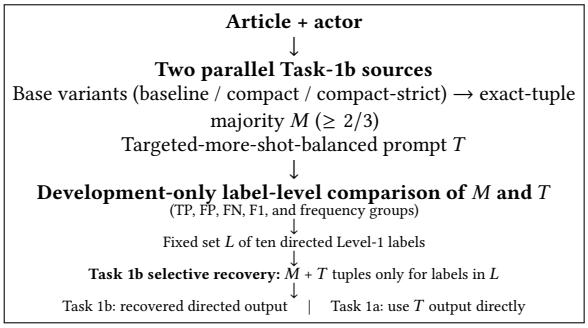



**Figure 1: WAIPS-01 uses a conservative exact-tuple majority base and a separate targeted prompt. Their development-set comparison fixes the labels eligible for Task 1b selective recovery; Task 1a uses the targeted output directly.**

We designed the prompt sequence to cover both sides of this trade-off. After reproducing the A0-style pipeline, we evaluated DeepSeek V4 Flash as a recall-oriented baseline: its Task-1b recall was .4845, compared with .2448 for GPT-4o A0. The compact variant removed the reasoning field, requested only strict JSON, and capped each actor at six values to reduce output cost and unnecessary explanation. Compact-strict then strengthened the evidence requirement to create a deliberately conservative member; it reached .6247 precision and .0180 DRR, but only .4255 recall. The later targeted-more-shot variants addressed the opposite failure mode with additional examples for difficult health, society, safety, recognition, reputation, and capability values. The balanced targeted variant was retained as the complementary source because it recovered more difficult labels without the most aggressive recall behavior of the earlier targeted prompt.

For an article–actor pair, each prompt returns a structured set of atomic Task-1b tuples $(a, v, d)$, where $a$ is the actor, $v$ is a Level-1 value, and $d$ is its aligned or contradictory direction. We use three DeepSeek V4 Flash variants to form a conservative base: (i) a baseline A0-style prompt, (ii) a compact prompt that requests concise structured JSON output, and (iii) a compact-strict prompt with a stronger evidence requirement. A fourth, targeted-more-shot-balanced prompt is designed as a complementary, higher recall source using targeted examples for difficult values. All prompt runs use structured Task-1 outputs and are evaluated with the official evaluator. This track also includes the end-to-end data, prompting, API, parsing, prediction, evaluation, and submission-validation pipeline. No model fine-tuning or learned calibration is used in WAIPS-01.

Figure 1 summarizes the complete procedure. The *Liu local* prompt used as one juror in FGHF is only one component from this work; it is not the complete label-aware method submitted as WAIPS-01, which includes conservative voting, label-level diagnosis, and selective addition.

### 3.2 Exact-tuple majority base

Let $B$, $C$, and $S$ denote the Task-1b prediction sets of the baseline, compact, and compact-strict variants. We retain an atomic tuple only when at least two of the three variants agree:

$$M = \{t : \mathbf{1}[t \in B] + \mathbf{1}[t \in C] + \mathbf{1}[t \in S] \geq 2\}. \quad (1)$$

The direction is part of $t$. Thus, two prompts that predict the same value with opposite directions do not support the same tuple. We deliberately use this precision-oriented exact-tuple rule instead

**Table 2: Directed labels eligible for selective recovery in WAIPS-01.**

| Direction | Level-1 value |
|---|---|
| Contradictory | Be creative |
| Contradictory | Have good health |
| Contradictory | Have a stable society |
| Contradictory | Have equality |
| Contradictory | Have harmony with nature |
| Aligned | Be capable |
| Aligned | Have good health |
| Aligned | Have a stable society |
| Aligned | Be humble |
| Aligned | Have a world at peace |

of a value-only vote followed by separate direction repair. It produces a stable base with development precision 0.6350 and DRR 0.0180, but its recall is only 0.4636 (Table 3).

The error counts show the same trade-off. Relative to the DeepSeek baseline, majority voting reduced Task-1b false positives from 868 to 637 (−231), while false negatives increased from 1,232 to 1,282 (+50). The effect was not uniform across the label distribution. Average per-label F1 changed only from .4796 to .4817 for frequent labels and from .2815 to .2866 for medium-frequency labels, but dropped from .2537 to .2168 for infrequent labels. These results support majority voting as a conservative base, but not as a complete method: its agreement rule disproportionately removed rare true positives. This pattern motivated the label-level diagnosis and selective recovery below.

## 3.3 Label-level diagnosis and selection rule

Aggregate F1 does not show *which* directed values are lost by the conservative base. We compared majority voting and the targeted prompt independently for every $(d, v)$ identity on the development set, recording TP, FP, FN, precision, recall, and F1. We also grouped labels by their frequency to check whether any aggregate gain was concentrated only on frequent classes. The diagnosis showed that majority voting made systematic false-negative errors on a small, interpretable group of values, while the targeted prompt improved their recall enough to offset its additional false positives.

We fixed the following ten directed Level-1 identities from this development comparison. The set is intentionally small: it limits selective addition to labels with observed complementary behavior rather than opening the final prediction set to every targeted-prompt output.

Let $T$ be the directed tuple set generated by the targeted-more-shot-balanced prompt and let $L_{\text{selected}}$ denote the fixed set in Table 2. The final Task-1b output is

$$P = M \cup \{t \in T : \text{label}(t) \in L_{\text{selected}}\}. \tag{2}$$

$M$ remains the default answer; $T$ is allowed to supplement it only for the chosen directed labels. This method is neither a blind union nor a weighted or calibrated vote, and it does not use fine-tuning. For Task 1a, we use the corresponding targeted-more-shot-balanced structured value output, as submitted in WAIPS-01.

## 3.4 Development results and analysis

Table 3 reports the trade-off that motivated the method. Relative to majority voting, selective recovery increases Task-1b Micro-F1 by 0.0298 and Macro-F1 by 0.0338. The gain is primarily recall: recall rises by 0.0602, compared with a precision decrease of 0.0201. DRR rises from 0.0180 to 0.0279, so recovering more directed values has a measured cost in direction reversals.

**Table 3: Task-1b development results for Liu's prompt-only variants. A dash denotes a metric that was not recorded for that run.**

| Method | Micro | Macro | Prec. | Recall | DRR |
|---|---|---|---|---|---|
| GPT-4o A0 | .3424 | .1496 | .5696 | .2448 | .0212 |
| DeepSeek baseline | .5245 | .3599 | .5716 | .4845 | .0243 |
| Compact | .5261 | .3511 | .5698 | .4887 | .0261 |
| Compact-strict | .5062 | .3234 | .6247 | .4255 | **.0180** |
| Majority vote | .5359 | .3532 | **.6350** | .4636 | **.0180** |
| Targeted prompt | .5387 | .3679 | .5750 | .5067 | .0302 |
| Label-aware ensemble | **.5657** | **.3870** | .6149 | **.5238** | .0279 |

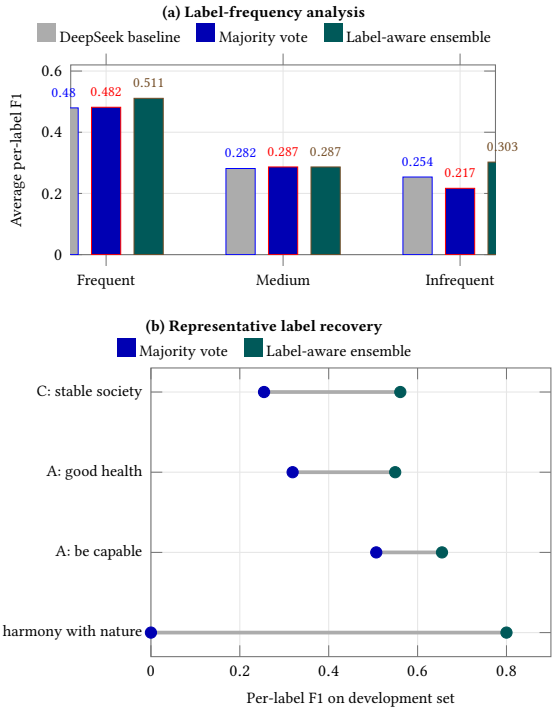



**Figure 2: Development-set evidence for diagnosis-guided selective recovery. (a) Majority voting slightly improves frequent and medium labels over the DeepSeek baseline but harms infrequent labels; selective recovery reverses that loss. (b) The method recovers selected directed labels without adding every targeted-prompt tuple; C and A denote contradictory and aligned, respectively.**

For Task 1a on development data, the DeepSeek baseline, compact, and compact-strict prompts obtained Micro/Macro-F1 of .6490/.5065, .6441/.4871, and .6125/.4576, respectively. The targeted prompt and the label-aware run both recorded .6555/.5044 because WAIPS-01 used the targeted structured output for Task 1a. The majority run recorded .6507 Micro-F1; its remaining Task-1a metrics were not retained in the experiment log.

The improvement extends beyond frequent labels. The average per-label F1 for the infrequent group rises from 0.2168 to 0.3025, whereas the medium group is unchanged (Figure 2(a)). Figure 2(b) shows representative cases. For example, contradictory *Have a stable society* improves from 0.2545 to 0.5611, aligned *Have good health* from 0.3188 to 0.5496, and contradictory *Have harmony with nature* from zero to 0.8000. These examples are consistent with the intended behavior: selective recovery addresses known gaps in the conservative ensemble rather than adding labels uniformly.

We also tested smaller subsets of the same ten labels using cached predictions, without additional LLM calls. Removing the low-impact label *aligned: Be humble* (ID 1:35; `small_best9`) gave .5656 Micro-F1, .3851 Macro-F1, .6150 precision, .5234 recall, and .0279 DRR. Removing *aligned: Have a stable society* (ID 1:27), the label most associated with direction errors (`small_low_drr`), gave .5613 Micro-F1, .3858 Macro-F1, .6222 precision, .5113 recall, and **.0230 DRR**. The .0230 result therefore comes from a final pre-submission development experiment, but it was a lower-DRR backup rather than the selected submission: the original ten-label setting retained the best Micro-F1 (.5657) with .0279 DRR.

### 3.5 Official result and limitation

WAIPS-01 was submitted as the direction-stable WAIPS run. On the official test, it achieved .6589 Micro-F1 and .5366 Macro-F1 on Task 1a, and .5543 Micro-F1, .3277 Macro-F1, and .0345 DRR on Task 1b. The last value is the lowest DRR among participant runs [1]. These results support using this run as a conservative alternative but do not establish dominance on every metric. The selection set was chosen on development data, and the development and official test sets contain different articles; consequently, the development comparison should not be read as a controlled test-time ablation.

The official precision/recall pairs were .6731/.6453 for Task 1a and .5991/.5157 for Task 1b. The submitted files contained 1,811 Task-1a labels and 2,058 directed Task-1b labels; 264 Task-1b predictions were selectively supplied by the targeted source. Before packaging, we verified complete article coverage, released actor membership, taxonomy validity, and the absence of duplicate tuples for the full development candidate; the corrected test submission used the same frozen construction and the required two-file Task-1 package structure. These checks reduce the likelihood that malformed output, rather than the intended label-aware pipeline, explains the reported result.

We treat the cross-split pattern as descriptive rather than as a controlled ablation. Task-1b Micro-F1 changed modestly from .5657 on development data to .5543 on the official test, whereas Macro-F1 fell from .3870 to .3277 and DRR rose from .0279 to .0345. This suggests weaker generalization for rare or difficult directed labels than for the aggregate prediction set. The lowest participant DRR indicates comparatively stable direction assignment, but it does not imply that rare-label recovery is solved. Because development and test contain different articles and gold distributions, these differences are descriptive evidence rather than estimates of a causal generalization gap [1].

## 4 Diagnosis-Guided Augmentation for Rare-Label Evidence

### 4.1 From label diagnosis to generation targets

Our initial DeBERTa classifier used a 108-dimensional sigmoid head and a single global threshold. Per-label analysis showed that this model had effectively learned no usable ordering for many low-frequency directed labels. We selected targets at the directed Level-1 class evaluated by Task 1b, rather than augmenting undirected values uniformly. A class was eligible when its official training count was at most 130 and its development support was at least three. Ranking the eligible classes by diagnostic evidence produced 18 frozen targets: 14 contradictory and four aligned classes, spanning 12 Level-2 groups.

The data were constructed in two rounds. Synthetic augmentation is widely used to improve sparse NLP supervision [11]; EAVIT also uses targeted data generation to strengthen underrepresented human-value labels [7]. Here we generate only for diagnosed directed FEHU labels rather than applying generic text perturbations. Round 1 covered six Level-1 values in both directions. DeepSeek V4 Pro generated 422 candidates; deterministic checks and an independent LLM semantic screen retained 340 silver records. These records were machine-screened, not manually verified. Round 2 concentrated on the 18 directed targets and allocated exactly 50 records to each target. Its 900 records mix targeted in-context generation, direction contrast cases, situation transformations, and hard negatives (Table 4). The hard-negative group includes 36 explicitly known negative labels. All Round-2 records passed schema, taxonomy, actor mention, direction, and duplicate checks, but they did not receive semantic LLM screening or human review. We report this provenance distinction because it limits what can be claimed about data quality.

**Table 4: Round-2 generation strategies.**

| Strategy | $n$ | Supervision and purpose |
|---|---|---|
| Targeted ICL | 450 | Full multi-label; varied contexts and co-values. |
| Direction contrast | 180 | Target-only; 90 aligned/contradictory pairs. |
| Transformation | 180 | Target-only; different actor, action, or setting. |
| Hard negative | 90 | Target-only; 36 explicit negative labels. |

Generation used explicit constraints rather than free-form paraphrasing. Each request fixed the actor, target identity, direction, and a news-style evidence requirement. Local validators rejected malformed mappings, absent actors, unsupported evidence sentences, forbidden meta-language, and duplicates before a record could enter the training data. The generation gates rejected and regenerated 67 Round-1 and 472 Round-2 candidates; the later Round-1 semantic screen separately excluded 82 of 422 records. Structural gates establish format and internal consistency; they do not show that every synthetic label is correct.

### 4.2 Partial-label training

The final training set contains 1,228 official articles, 340 Round-1 records, and 900 Round-2 records: 2,468 articles and 4,575 actor examples in total. The synthetic news vignettes are often only partially annotated. This matches the partial multi-label setting in which unspecified labels remain unknown rather than being forced negative [12]. Treating every unlisted class as negative would create hundreds of false-negative training signals, especially for the single-target records. We mark 3,785 examples as fully supervised and 790 as target-only. For a target-only example, binary cross entropy is evaluated only on its known positive entries and any explicitly known negatives; the remaining label positions receive a zero loss mask.

We fine-tune DeBERTa-v3-base [13] with LoRA adapters [14] on the query and value projections ($r = 16$, $\alpha = 32$, dropout 0.1) and a 108-way classification head. Each input pairs the article text with the actor name and uses a maximum length of 512. We optimize the model with class-weighted binary cross entropy. The positive weight for each class is $\sqrt{n_{\text{negative}}/n_{\text{positive}}}$, clipped to $[1, 10]$ and computed over supervised entries only. We select checkpoints by development Task-1b Micro-F1 and decode predictions at a global threshold of 0.30. In FGHF, the encoder never assigns the final direction: we export all 108 sigmoid probabilities and use them only as continuous jury features.

**Table 5: Score-level tail diagnostics on development data.**

| Diagnostic | Official | Augmented v2 |
|---|---|---|
| Mean AUC (18 targets) | .536 | **.697** |
| Mean oracle F1 | .053 | **.112** |
| AUC ≤ .55 | 11/18 | **3/18** |

**Table 6: Three-seed target analysis (mean ± SD).**

| Training condition | AUC | Oracle F1 |
|---|---|---|
| Official only | .513 ± .030 | .048 ± .010 |
| +900, assume negative | **.624 ± .011** | **.095 ± .014** |
| +900, masked | .577 ± .028 | .080 ± .019 |
| +340+900, masked | .632 ± .105 | .111 ± .049 |

For example $i$ and directed class $k$, let $m_{ik}$ be one for an observed label decision and zero for an unknown entry. Training minimizes

$$\mathcal{L} = \frac{\sum_{i,k} m_{ik} w_k \, \mathrm{BCE}(y_{ik}, z_{ik})}{\sum_{i,k} m_{ik}}. \tag{3}$$

All mask entries are one for official and fully labelled records. On target-only records, only the intended positives and explicitly supplied hard negatives contribute. This does not rebalance the dataset to 50:50; it prevents missing synthetic annotations from becoming certain negatives. The distinction matters in a 108-class problem, where a single partially annotated example would otherwise create more than one hundred negative loss terms.

## 4.3 Effect of augmentation on rare-label evidence

On the 18 targets, mean target AUC with official data only was 0.536, and mean per-target oracle F1 was 0.053. The augmented v2 checkpoint raised these values to 0.697 and 0.112; AUC improved for 14 of 18 targets and oracle F1 for 16 of 18. Thus the generated supervision changed the ranking of target positives, rather than merely shifting an unchanged distribution across the 0.30 threshold.

The three-seed comparison, which isolates the 900 Round-2 records, shows the same pattern: the official-only condition remains close to chance (mean target AUC 0.513), whereas the 900 records raise AUC to 0.624 with assume negative supervision and 0.577 with masking. Both improvements have the same sign for all three matched seeds. With the complete 340+900 data, masked supervision was retained because it had the better mean full-data result and protected the sparsely annotated Round-1 records; its mean target AUC was 0.632, but with a standard deviation of 0.105.

The gain does not establish a reliable tail classifier. Three full-data seeds showed substantial variance: target true positives were 0, 0, and 37, and overall Macro-F1 ranged from 0.022 to 0.182. Given this variance, the encoder is used as an auxiliary source rather than a submitted standalone model. The clearest contribution of augmentation is improved low-frequency score separation; the resulting DeBERTa probabilities then supply continuous auxiliary evidence to FGHF. Together with E6b's direct tail recall and the gated fusion stages, this is one contributor to the stronger macro performance of the final system.

**Table 7: Development profiles and assigned FGHF roles. "Liu local prompt" denotes one targeted juror, not the complete label-aware ensemble submitted as WAIPS-01.**

| Member | Micro/Macro | DRR | FGHF role |
|---|---|---|---|
| E1 | .5480/.3719 | .0347 | Balanced juror |
| E6 | .5412/.4079 | .0504 | Recall juror |
| **E6b** | **.5496/.4124** | .0459 | Primary generator |
| E6c | .5343/.3725 | .0657 | Value/disagreement juror |
| Liu local prompt | .5347/.3385 | **.0230** | Precision juror |
| DeBERTa | — | — | Continuous evidence |

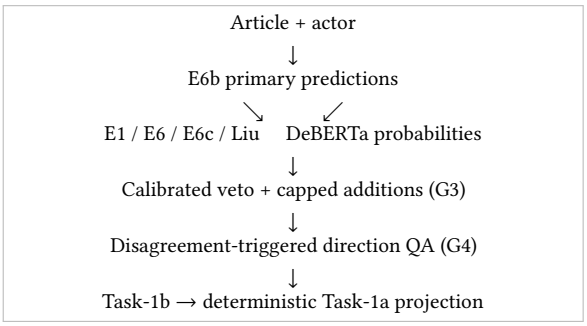



**Figure 3: FGHF separates value generation, calibrated fusion, and direction repair.**

# 5 Frequency-Gated Heterogeneous Fusion (FGHF)

## 5.1 Members with deliberately different roles

FGHF starts from E6b, a prompt pipeline that first segments an article into six to nine coherent developments and then predicts values for each actor using a fixed few-shot scaffold. Although limited to segmentation granularity, this change affected development performance. Its predecessor E6 produced a median of 12 segments and achieved high recall with many false positives. Asking the segmenter to merge closely related sentences reduced the median to eight. On development data, E6b improved over E6 from 0.5412 to 0.5496 Micro-F1 and from 0.4079 to 0.4124 Macro-F1. Relative to the earlier E1 prompt run, its Micro-F1 changed only slightly (0.5480 to 0.5496), but Macro-F1 rose by 0.0405. We therefore use E6b as the primary generator because it provides broad coverage that a later fusion stage cannot supply on its own. This use of localized event units is consistent with NEVU's event-centric, multi-granularity formulation [2]; E6b generates its own developments from each article at inference time.

The jurors serve different roles. E1 uses a balanced article-level prompt, whereas E6 uses finer self-segmentation to favor recall. E6c runs DeepSeek Pro without thinking over the E6b segments. It has the strongest individual Task-1a development result (0.6774) but weak direction behavior; we use it as value evidence, not as a direction authority. The local Liu component is one targeted balanced few-shot prompt run and is distinct from the complete label-aware method submitted as WAIPS-01 in Section 3. The augmented DeBERTa model contributes continuous probabilities rather than discrete predictions.

## 5.2 Calibrated veto and additions

For an actor and directed value cell $c$, the removal score combines exact-direction votes, value-only votes, and the encoder's directionless value score:

$$s_{\mathrm{rm}}(c) = n_{\mathrm{exact}}(c) + 0.25 n_{\mathrm{value}}(c) + p_D(\mathrm{value}(c)). \tag{4}$$

For a candidate not emitted by E6b, the addition score is stricter:

$$s_{\text{add}}(c) = n_{\text{exact}}(c) + p_D(c). \tag{5}$$

Pool-adjacent-violators calibration maps each monotone score to an empirical probability of being gold. During development, calibration is two-fold cross-fitted by article, so an article's own labels do not determine its mapping. This monotone score-calibration approach follows the established use of isotonic regression for classifier probabilities [15]. For test inference, the mapping is fitted once on the full development set and applied unchanged. Following the calibrated-probability F1 result of Lipton et al. [16], we use the plug-in threshold $\tau = F1_{\text{primary}}/2$; the frozen full-development value is 0.27479.

The rules are intentionally asymmetric. A primary cell is removed only when its calibrated probability is below $\tau$, it belongs to one of 38 frequent directed classes (official training count greater than 130), and the actor retains at least one label. A non-primary cell can be added only when it was proposed by a juror, its calibrated probability reaches $\tau$, it is in the same frequent tier, and E6b does not already hold the value in either direction. Additions are capped at two per actor. In the ungated development control, removing weak tail predictions reduced Macro-F1 sharply. The frequency gate also means that the 18 augmentation targets are not directly switched on or off by the G3 actions; the encoder supplies broader score evidence, while E6b remains the main direct source of tail labels.

### 5.3 Focused direction repair

The last stage addresses direction errors separately. A fused prediction is sent to direction QA only if at least one juror predicts the same value in the opposite direction and the fused set does not already contain both directions. DeepSeek Flash then receives the article, actor, and one value and returns only `aligned` or `contradictory`. On development data, 176 cells were flagged and 46 were flipped; seven failed all retries and were retained with explicit failure records. On test, 207 cells were flagged, 77 were flipped, and five failed and were retained. Because this stage never introduces a new value, it can repair direction without reopening the value-selection problem.

### 5.4 Frozen development-to-test transfer

The complete test path was executed as a fit–apply procedure. All prompt members read only the official Task-1 test article file; E6 and E6b generated their own segmentations because official test subevents were unavailable. The removal and addition calibrators were fitted on all development data and loaded unchanged for test. We also froze the frequency tier, score weights, threshold, maximum of two additions, and QA trigger. Every empty API response or parsing failure was retried. If retries were exhausted, the failure either stopped the stage or, under the predeclared QA behavior, was logged while retaining the original direction.

Before submission, both files were checked for complete article coverage, valid actor membership, taxonomy ranges, duplicate tuples, and deterministic Task-1a projection. The final WAIPS-03 output contained 2,764 directed labels, covering 93 of the 108 classes and all 54 Level-1 values. These counts cannot substitute for test labels, but the removal/addition profile was close to development and gave no operational indication of a broken transfer pipeline.

## 6 Experiments and Results

### 6.1 Development results

Table 8 reports the successive stages of FGHF. E6b already raises Macro-F1 by 0.0405 over E1. This initial gain matters when interpreting the final macro-F1 result. The calibrated veto (G1) mainly restores precision. Re-enabling supported additions (G3) provides the largest single fusion increment, and direction QA (G4) produces a smaller but consistent improvement in Micro-F1, Macro-F1, and DRR.

**Table 8: Development results. T18 is Micro-F1 over the 18 augmentation targets.**

| System | 1b Micro | Precision / Recall | 1b Macro | 1a Micro | DRR | T18 |
|---|---|---|---|---|---|---|
| E1 | .5480 | .541 / .556 | .3719 | .6574 | .0347 | .2460 |
| E6b primary | .5496 | .502 / .608 | .4124 | .6621 | .0459 | .3145 |
| G1: calibrated veto | .5608 | .559 / .563 | .4146 | .6587 | **.0374** | .3145 |
| G3: veto + additions | .5790 | .539 / .626 | .4266 | .6874 | .0450 | .3145 |
| **G4: G3 + direction QA** | **.5817** | **.542 / .629** | **.4308** | **.6874** | .0419 | **.3264** |

The G4–E6b Micro-F1 difference is 0.0321. In a paired article bootstrap with 10,000 resamples, the 95% interval runs from 0.0209 to 0.0433. This interval measures article-sampling variation; it does not measure nondeterminism across LLM calls. G1 removed 484 predictions on development data and G3 admitted 364. The test pipeline was frozen before inference and produced a similar action profile: 529 removals and 365 additions. These counts are useful for checking operational consistency, not for estimating test-set quality.

Two controls help explain the final gated rule. Removing the local Liu prompt from the veto jury changes Micro-F1 from .5608 to .5619. This control concerns only that single FGHF juror; it is not an ablation of Liu's complete label-aware ensemble submitted as WAIPS-01 in Section 3. The juror's precision-oriented profile is more relevant when additions are considered. Conversely, allowing vetoes on all classes gives 0.5703 Micro-F1 but only 0.3900 Macro-F1. This is the failure mode the frequency gate was designed to prevent: a weak auxiliary score can appear favorable in aggregate by suppressing difficult labels that macro-F1 is intended to expose.

### 6.2 Where the macro gain enters

The development results localize the Macro-F1 gains more precisely than the final-run comparison alone. Moving from E1 to E6b raises Macro-F1 from 0.3719 to 0.4124 (+0.0405) while changing Micro-F1 by only +0.0016. The frequent-class veto then adds +0.0022 Macro-F1, supported additions add a further +0.0120, and direction QA adds +0.0042. The 18-target Micro-F1 is 0.3145 for E6b and remains unchanged through G1 and G3; direction QA raises it to 0.3264.

This decomposition is consistent with the assigned roles. E6b directly emits most of the additional tail labels. Augmentation improves rare-label separation in the DeBERTa score space and contributes a continuous confidence signal, but the frequent-class gate prevents G3 from directly adding the 18 selected targets. Its support for the final macro-oriented system is therefore indirect and shared with the prompt jury. We regard the official macro advantage as consistent with the combined design, not as a demonstration that augmented DeBERTa alone caused it.

### 6.3 Official results

The Task Overview [1] reports four evaluated run packages from two teams: three from WAIPS and one from IIUoT. The results

**Table 9: Official Task-1 results. Each F1 entry is Micro / Macro. The organizer baseline is unranked; bold marks the best participant.**

| System / run | Task 1a F1 | Task 1b F1 | DRR |
|---|---|---|---|
| Organizer baseline | .6782 / .5405 | .6284 / .4402 | .0386 |
| IIUoT-01 | **.7042 / .5808** | **.5979** / .3930 | .0613 |
| WAIPS-01 | .6589 / .5366 | .5543 / .3277 | **.0345** |
| WAIPS-02 | .6930 / .5646 | .5496 / .3580 | .0395 |
| **WAIPS-03** | .6862 / .5705 | .5808 / **.3952** | .0477 |

show different trade-offs rather than one uniformly dominant run. WAIPS-02 gives our highest Task-1a Micro-F1, WAIPS-03 gives our strongest Task-1b Micro- and Macro-F1, and WAIPS-01 gives the lowest DRR among participant runs. Across teams, IIUoT-01 leads the primary Micro-F1 ranking on Tasks 1a and 1b, while WAIPS-03 has the highest participant Task-1b Macro-F1. The organizer baseline is an unranked reference and exceeds every participant run on Task-1b Micro- and Macro-F1. We state the macro result as a participant comparison, not as overall task dominance.

The development and official results show the same broad, though not fully uniform, pattern. Broad value recovery helps macro-F1, but it increases exposure to direction mistakes. WAIPS-01 is the conservative end of this spectrum. FGHF accepts a higher DRR in exchange for more labels and higher Task-1b F1. Its targeted direction QA reduces the development reversal rate after additions, but does not erase the underlying coverage–direction trade-off on the official test set.

### 6.4 Error patterns and limitations

We observed two recurring error types. The first is over-generation of plausible values for a nearby actor or for the article in general. Self-segmentation makes more local evidence visible, but a value can still be attached to an actor who is mentioned rather than responsible. The calibrated veto repairs some of these errors only in frequent classes. The second is direction ambiguity. Jurors often agree that a value is relevant while disagreeing about whether the actor supports or undermines it; focused QA helps, but ambiguous reporting and multiple actions by one actor remain difficult.

Several limitations affect this interpretation. The fusion weights, frequency gate, and threshold family were designed through analysis of development data, even though calibration was cross-fitted and the test pipeline was frozen. The Round-2 synthetic data were structurally checked but not semantically screened, and masked DeBERTa training has high seed variance. The augmentation experiments show improved low-frequency score separation, and the augmented encoder provides auxiliary evidence to the fusion. We therefore treat it as one contributor to the higher Macro-F1 rather than the sole source of the gain; E6b and the LLM jury remain essential. Results from two participating teams should not be read as a broad leaderboard conclusion.

## 7 Related Work

**Computational human-value recognition.** The FEHU taxonomy ultimately builds on the refined theory of basic individual values [3]. Human-value identification was studied earlier in arguments [4, 5], where transformer ensembles were also competitive [6]. EAVIT later combined a local value detector with an online LLM and used targeted generation for underrepresented values [7]. NEVU introduced actor-conditioned, direction-aware recognition at multiple event granularities in factual news [2]; FEHU turns this setting into shared-task subtasks with an official evaluation protocol [1].

**Event and actor grounding in news.** NEVU's multi-granularity formulation provides the closest precedent for using localized event context in actor-conditioned value recognition [2]. Our E6b pipeline generates its own compact developments at inference time rather than relying on annotated subevents, while FEHU fixes the actors and evaluation units.

**Ensembling, imbalance, and calibrated auxiliary models.** Consistency-based prompting can improve reliability by aggregating multiple reasoning paths [8], and the Adam-Smith ValueEval system showed that model ensembling can also be effective for human-value classification [6]. Liu's label-aware method differs by voting on exact structured tuples and permitting selective recovery only for directed labels diagnosed on development data. This restriction is motivated by long-tail learning [9, 10]. Our second methodological line uses targeted synthetic data [7, 11], partial-label supervision [12], and a DeBERTaV3/LoRA auxiliary encoder [13, 14]. FGHF calibrates monotone ensemble scores with isotonic regression [15]. The FEHU-specific prompt designs, directed-label diagnostics, and gate configurations were developed in this project.

## 8 Conclusion

Our Task-1 systems were built by treating label recovery, precision control, and direction assignment as separate problems. Liu's label-aware method, submitted as WAIPS-01, uses label-level diagnosis to recover selected misses without abandoning a conservative prompt majority. The augmentation track gives a LoRA-tuned DeBERTa encoder substantially better score-level separation on selected rare directed labels and supplies auxiliary evidence, while E6b provides strong direct tail recall through granularity-controlled self-segmentation. FGHF combines these heterogeneous signals with calibrated, frequency-gated veto and addition rules and repairs direction only where jurors disagree. The official results reflect the intended trade-off: WAIPS-03 reaches 0.5808 Micro-F1 and 0.3952 Macro-F1 on Task 1b, while WAIPS-01 remains more direction-stable. More stable tail training remains an important direction for future work.

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